

Mariam Ali Akbar

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SUMMARY

Computer Science graduate with hands-on experience in web development, full-stack applications, AI/RAG systems, API development, HubSpot automation, and CRM data workflows. Experienced with JavaScript, TypeScript, React, Next.js, PostgreSQL, Prisma, and REST APIs, with a focus on building practical applications and learning through independent projects.

EXPERIENCE

Software Engineering Intern | [Current Company] | 2026

- Built and reviewed web development projects including a Movie Explorer application using Next.js and TypeScript and a forms project using React Hook Form, Zod, Prisma, and PostgreSQL.
- Developed an API to create HubSpot workflows through Claude Code when the available HubSpot MCP functionality did not support workflow creation.
- Worked with HubSpot workflows, data enrichment, CRM integrations, orphan company/contact identification, and Metadata.io forms and metadata.
- Worked with Google Tag Manager (GTM) and Google Analytics and reviewed Metadata.io tags as part of digital marketing and analytics tasks.

Hybrid Intern | Punjab Information Technology Board (PITB), Lahore | 2026

- Developed backend APIs using Express.js for CRUD operations, routing, file uploads, and JWT-based authentication.
- Tested REST APIs using Postman and integrated relational data using MySQL.

Frontend Intern | Bytewise Limited | 2024

- Developed responsive web interfaces using HTML, Tailwind CSS, and JavaScript.
- Built interactive React.js projects including Movie Explorer, Expense Tracker, Shopping Cart, and To-Do applications.
- Practiced component-based architecture, Git version control, and Vercel deployment.

PROJECTS

Movie Explorer | Next.js, TypeScript, TMDB API, Tailwind CSS, shadcn/ui | Personal Project

- Built a movie discovery application with trending, popular, top-rated, upcoming, and now-playing sections, dynamic movie pages, search, favorites, and recently viewed movies.
- Implemented debounced movie search with URL-based search state and keyboard-accessible search interactions.
- Improved the interface with reusable components and shadcn/ui while deploying the application with Vercel.

WordKnit | React, Node.js, Express.js, MongoDB, Python, Docker, RAG, LLM APIs | Final Year Project

- Built a full-stack vocabulary and reading companion with AI-powered contextual word explanations and speaking/vocabulary coaching features.
- Implemented a RAG pipeline using TF-IDF vectorization and PyMuPDF to retrieve contextual definitions from vocabulary resources.
- Integrated an LLM to generate contextual word explanations, synonym nuances, and memory-oriented learning support.
- Built document/PDF and OCR features using PyMuPDF, Tesseract OCR, OpenCV, and PDF.js.

RAG-Based Document Chatbot | Next.js, Node.js, PostgreSQL, Vector Search, LLM | Project

- Developed a retrieval-augmented generation (RAG) document chatbot that allows users to ask questions about uploaded documents and receive context-grounded answers.
- Implemented document processing, text chunking, embeddings, semantic retrieval, and LLM-based response generation.
- Displayed retrieved document sources to help users trace answers back to the uploaded content.

E-Commerce Application | React.js, JavaScript | Personal Project

- Built an e-commerce application to strengthen understanding of React components, state management, and React hooks.

EDUCATION

BS Computer Science | University of Okara, Renala Khurd | 2022–2026

- CGPA: 3.7/4.0

CERTIFICATIONS & LEARNING

- CS50x — Harvard / edX
- Data Structures & Algorithms, Database Systems, PostgreSQL, React, Next.js, TypeScript

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C++, SQL

Frontend: React.js, Next.js, HTML, CSS, Tailwind CSS, shadcn/ui

Backend: Node.js, Express.js, REST APIs, JWT Authentication

Databases: PostgreSQL, MongoDB, MySQL, Prisma

AI: RAG, LLM Integration, TF-IDF, OCR

Tools & Platforms: Git, GitHub, Docker, Vercel, Postman, HubSpot, Google Tag Manager, Google Analytics